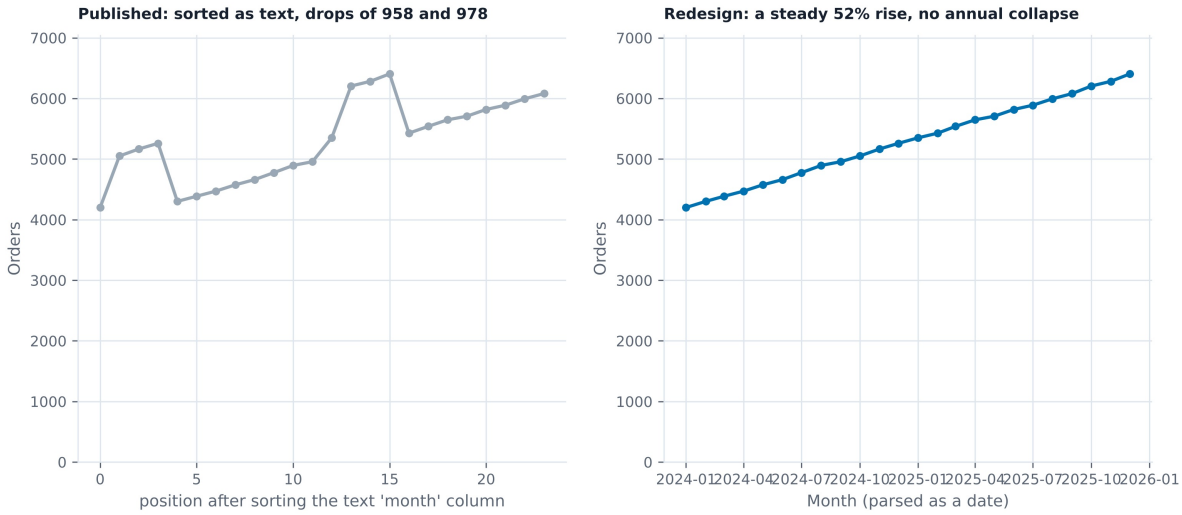
Part A — Diagnosis of the nine published charts

Each case has a figure, `partA_caseNN.png`: the left panel reproduces the published recipe, the right panel is the redesign. Every number below comes from `partA_charts.py` (printed in `partA_selfcheck.txt`). Data is the synthetic case file, CDB601220.

Format note: the brief asks for a PDF. This diagnosis is written as markdown with the figures as PNG, and the PNGs can be exported to PDF on request.

Case 1 — Orders by month, string dates

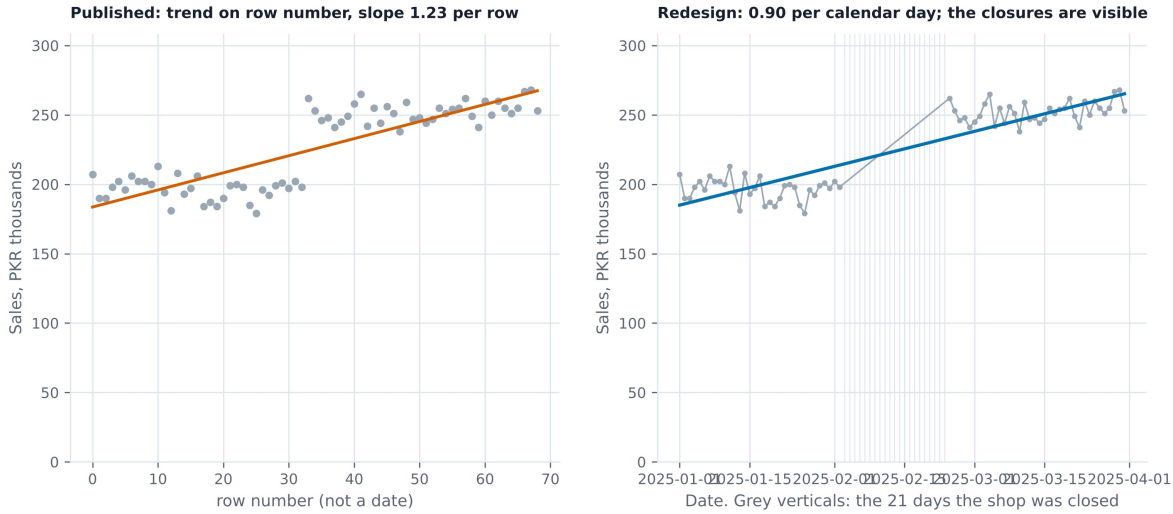


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: '2024-1' sorts before '2024-2' as text, so sorting by the string puts Jan 2025 after Dec 2024 for the wrong reason. Pick the date, not the label.

Case 1: published recipe (left) and redesign (right)

- **Conclusion:** the published series shows a collapse that does not exist.
- **Decoy:** the sawtooth drops that look like a seasonal cycle.
- **Mechanism:** months are stored as text (“2024-1”, “2024-2”). Sorting the text puts “2024-10” and “2024-11” in the wrong place, so two December-to-January transitions come out as drops of 958 and 978 orders.
- **Number:** parsing the month as a date gives a steady rise from 4,200 to 6,405 orders, +52.5%.
- **Redesign:** parse the month to a date before sorting. Plot the parsed series on a date axis.

Case 2 — Daily sales with closed days

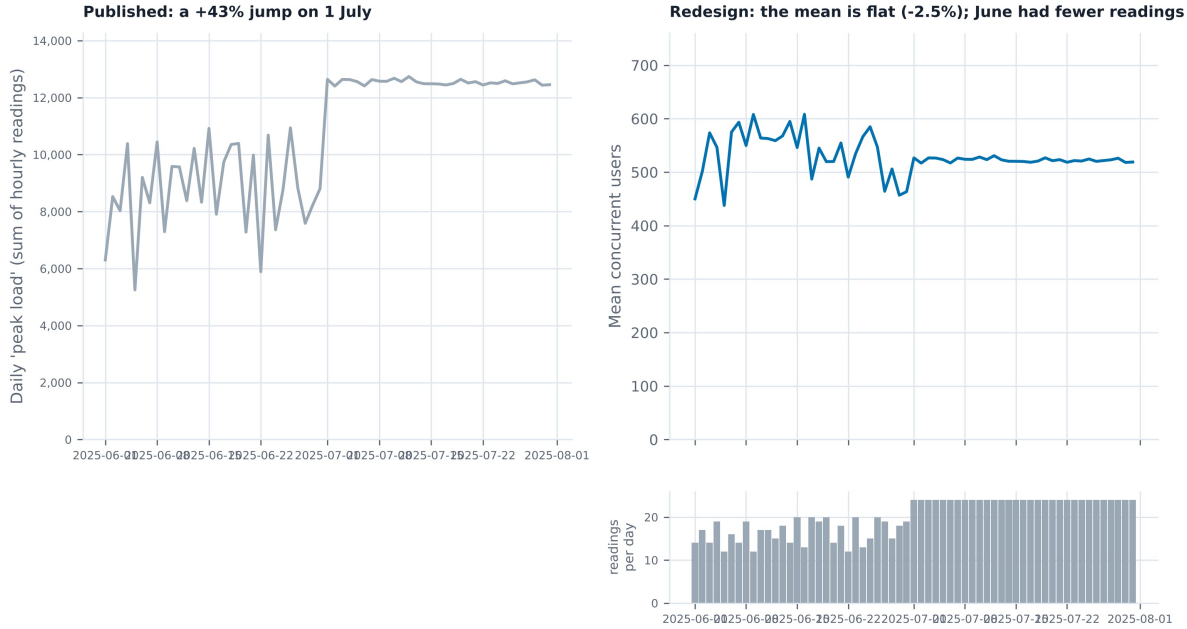


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: the x axis counts recorded rows, so each closed day is silently removed and the 21 closures pull the days together. The row slope is inflated by 37%.

Case 2: published recipe (left) and redesign (right)

- **Conclusion:** the published trend slope is overstated, and the closures are hidden.
- **Decoy:** a clean-looking linear trend on a row-number axis.
- **Mechanism:** the x-axis counts recorded rows. The 21 closed days (three per weekday) are removed silently, so the axis compresses time. The row slope is 1.232 per row; the calendar-day slope is 0.901 per day. The row axis overstates the slope by about 37%.
- **Number:** first to last recorded value 207 to 253, +22.2%.
- **Redesign:** plot against calendar date and show the closed days as gaps or verticals.

Case 3 — Bandwidth as hourly concurrent users

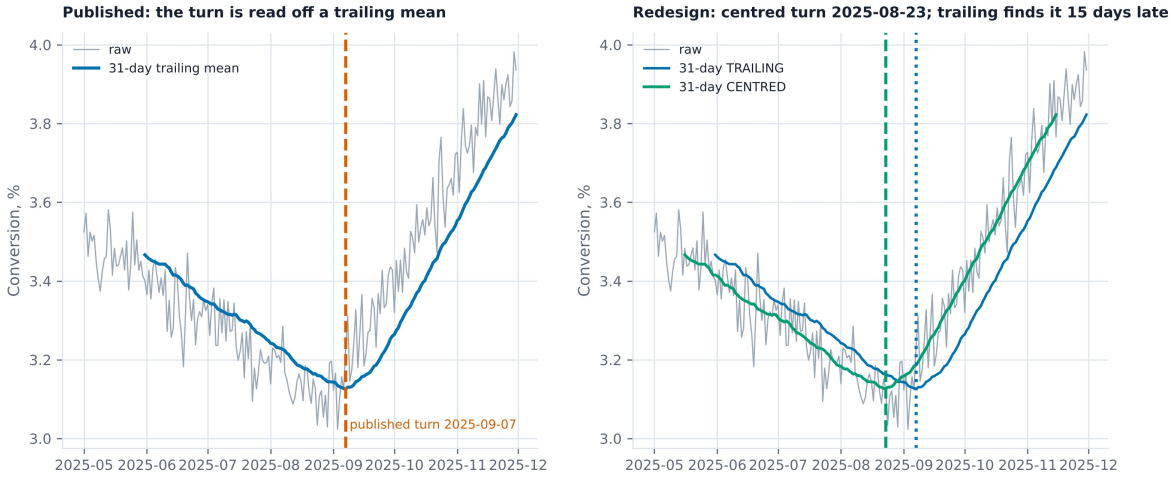


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: concurrent_users is a level. June logs 12-20 readings a day and July 24, so the daily SUM rises with the number of readings, not with users.

Case 3: published recipe (left) and redesign (right)

- **Conclusion:** the +43% July jump is a counting artefact. The average load is flat.
- **Decoy:** a sudden step on 1 July in the daily total.
- **Mechanism:** `concurrent_users` is a level (how many users are connected at one moment), not a flow. June has 12–20 readings per day and July has 24, so summing readings per day counts more readings in July.
- **Number:** daily sum June mean 8,782, July mean 12,543, +42.8%. Daily mean 535.9 to 522.6, −2.5%.
- **Redesign:** plot the daily mean, and show readings-per-day in a panel beneath it so the coverage change is visible.

Case 4 — Intervention and conversion

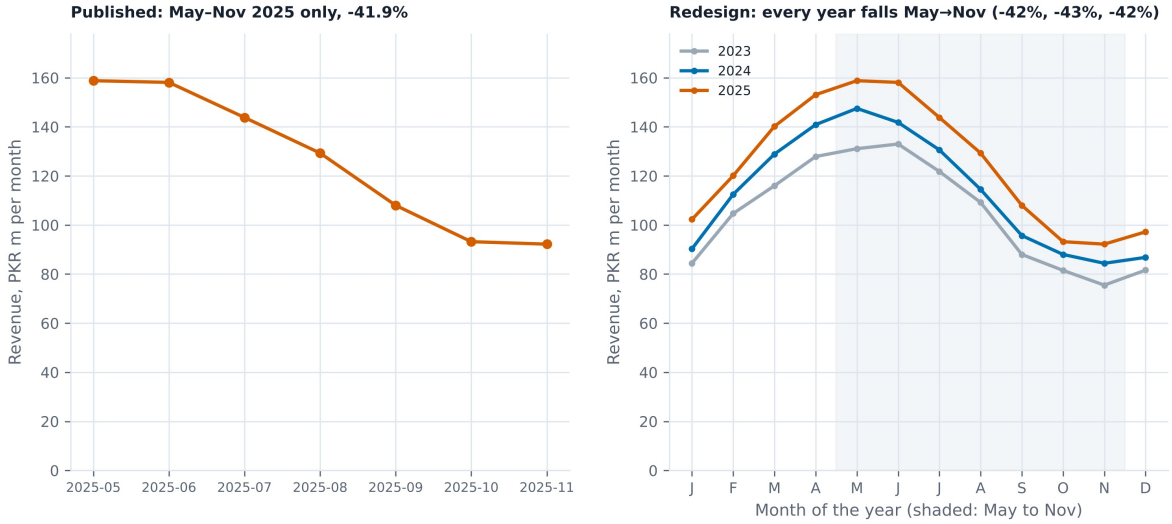


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: a trailing window cannot see a turn until it has filled, so its minimum sits half a window (15 days) after the true turn.

Case 4: published recipe (left) and redesign (right)

- **Conclusion:** the turn is in mid-August, not early September.
- **Decoy:** a turn read off a trailing 31-day minimum.
- **Mechanism:** a trailing window cannot see a turn until it has filled, so its minimum falls half a window after the true turn.
- **Number:** trailing minimum on 7 Sep 2025; centred minimum on 23 Aug 2025. Lag = 15 days, which equals half of 31.
- **Redesign:** use a centred window for locating the turn, and say in the caption that the trailing version is for live monitoring only.

Case 5 — Revenue, May to November 2025

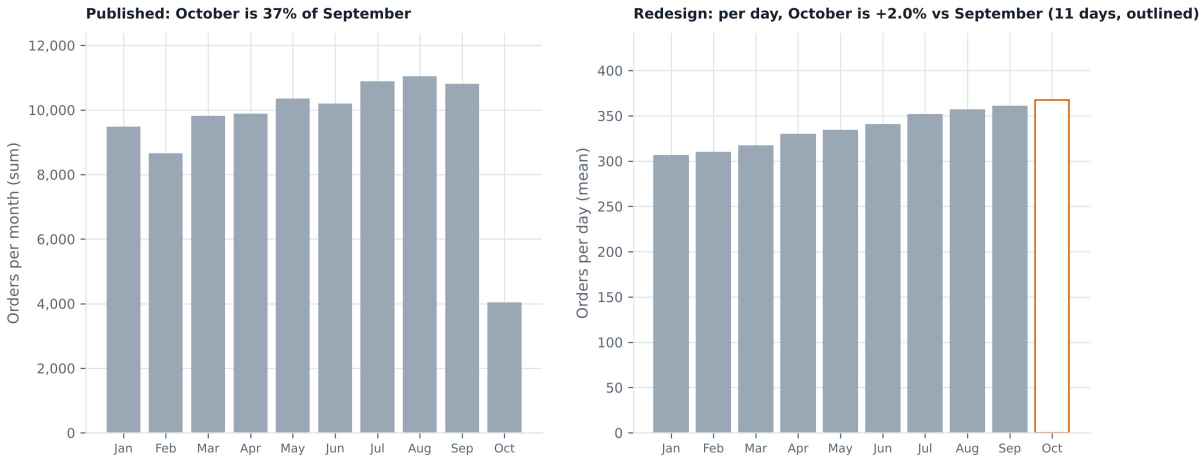


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: a seasonal peak in May or June followed by a fall to November is the pattern of every year in the file. The window starts at the peak.

Case 5: published recipe (left) and redesign (right)

- **Conclusion:** the May-to-November fall is seasonal, not a 2025 decline.
- **Decoy:** a −41.9% fall in the window shown.
- **Mechanism:** the window starts at the seasonal peak. Every year in the file falls by the same amount from May to November.
- **Number:** 2023 −42.4%, 2024 −42.8%, 2025 −41.9%. November 2025 is 9.3% above November 2024.
- **Redesign:** plot each year on a month-of-year axis and shade May to November, so the repeated pattern is visible.

Case 6 — Monthly orders, file ends on 11 October

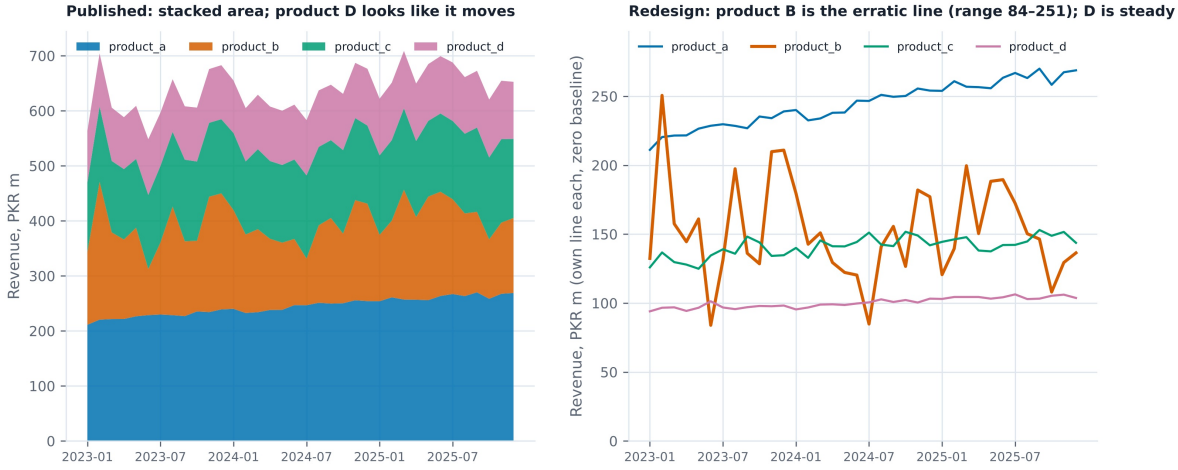


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: the file stops on 11 October. A month summed over 11 days is compared with a month summed over 30.

Case 6: published recipe (left) and redesign (right)

- **Conclusion:** October looks like a collapse; per day it is about flat.
- **Decoy:** October at 37% of September.
- **Mechanism:** the file stops on 11 October. October is summed over 11 days and September over 30.
- **Number:** October 4,041 over 11 days (367 per day). September 10,805 over 30 days (360 per day). Per day, +2.0%.
- **Redesign:** compare means per day and outline the partial month.

Case 7 — Stacked area of four products

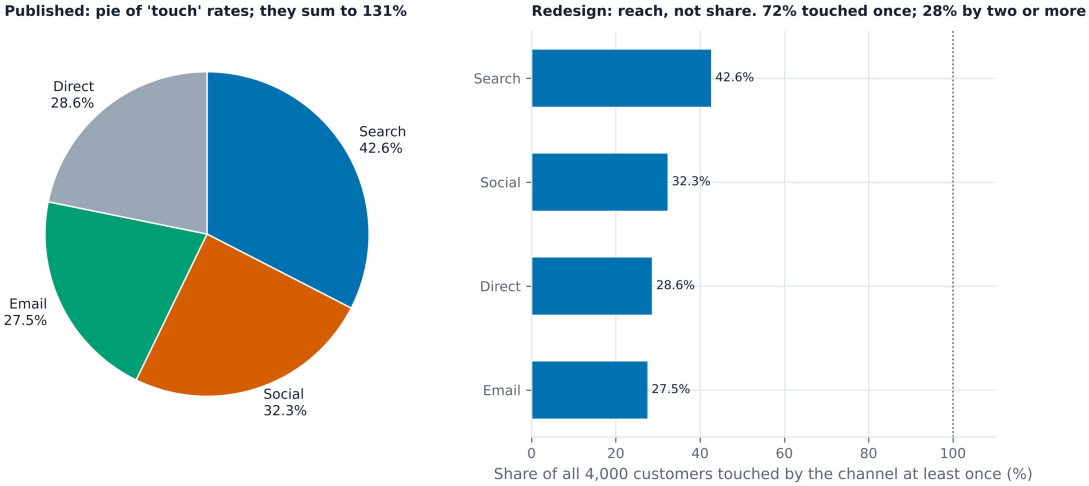


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: in a stack, every band's edges move with everything beneath it, so product B's swings are read as D's.

Case 7: published recipe (left) and redesign (right)

- **Conclusion:** product B is the erratic series; D appears to move only because of the stack.
- **Decoy:** the top band of the stack looks as if it is swinging.
- **Mechanism:** in a stacked area, each band's edges shift with every band beneath it.
- **Number:** product B ranges 83.9–250.6 (SD 35.6). Product D ranges 94.0–106.3 (SD 3.6). The total stays inside 548.3–708.7, which matches the brief.
- **Redesign:** plot each product as its own line on a zero baseline, with B highlighted.

Case 8 — Channel attribution as a pie

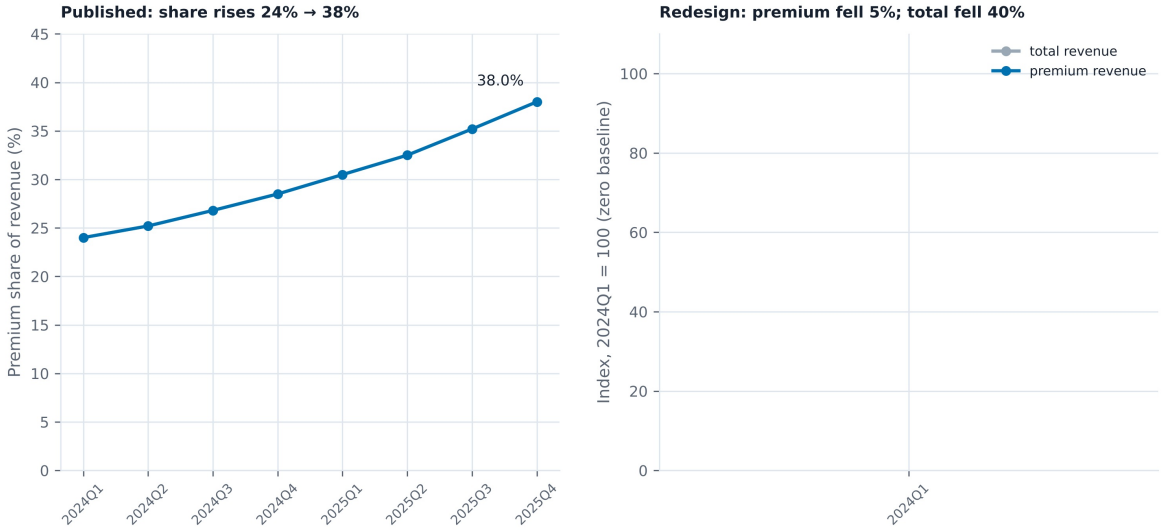


Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: a customer can be touched by several channels, so the four rates are reach and overlap. A pie assumes one channel per customer.

Case 8: published recipe (left) and redesign (right)

- **Conclusion:** the four channel rates are not shares of customers; they are overlapping reach.
- **Decoy:** a pie that sums to 131%.
- **Mechanism:** a customer can be touched by several channels. 2,895 customers have one touch, 980 have two, 116 have three, 9 have four. Zero customers have no touch.
- **Number:** search 42.6%, social 32.3%, direct 28.6%, email 27.5%. Sum 131.0%. 72.4% touched once; 27.6% touched two or more times.
- **Redesign:** reach bars with a 100% reference, plus the exclusive-combination counts.

Case 9 — Premium share of revenue



Source: Week 4 Part A case file (synthetic, CDB601220). Left: the published recipe. Right: the redesign. Mechanism: the share's denominator shrank faster than the numerator. Premium revenue barely moved; the rest of the business fell.

Case 9: published recipe (left) and redesign (right)

- **Conclusion:** the premium line did not grow; the denominator shrank.
- **Decoy:** share rising from 24% to 38%.
- **Mechanism:** total revenue fell from 1,000 to 600. Premium revenue went from 240 to 228, a fall of 5%. The share rose because the rest of the business fell faster.
- **Number:** premium share 24.0% to 38.0%. Premium −5.0%; total −40.0%.
- **Redesign:** plot absolute premium revenue, and index total and premium to 2024Q1 = 100 on a zero baseline.

Open points

- Case 3's redesign top-left panel is the published recipe only; the readings-per-day panel sits on the right, so the left side has no bottom panel. This is deliberate.
- Case 6's October outline is a visual cue, not a computed value; the per-day numbers are the evidence.